Introduction to Data Analytics
MS - 4610

Session 11 – Classification: Logistic, LDA and SVMs

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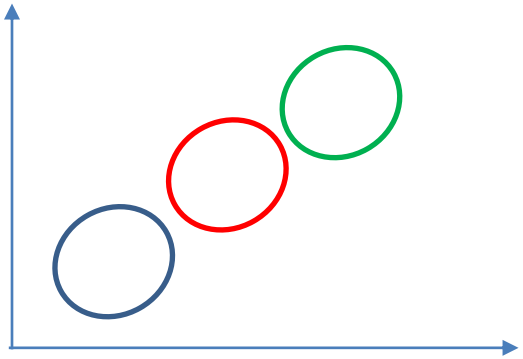
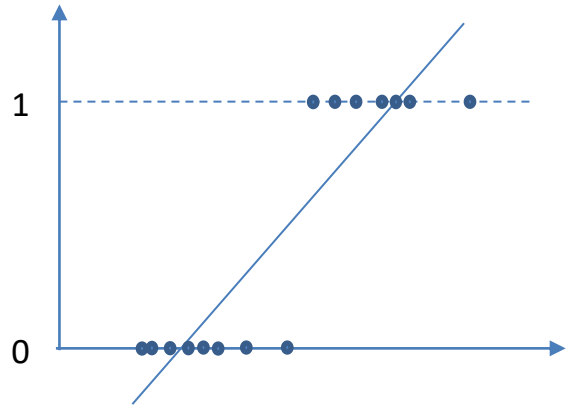
Classification Techniques

- Major concepts
  - What are classification problems? How is it different from regression
  - Why is it sometimes confused with Clustering?
  - What does complexity in classification mean?
  - What classifier have we already seen?
  - What is the problem of using regression to classify
    - Coding approach for 2 classes (works OK, but assumptions?)
    - More than two classes:
      - Simple coding
      - The 0 and 1 approach for each class. Where does it fail?



Logistic Regression

- Take the binary output case:
  - What is the distribution of residuals?
  - What about the scatterplot to start off the visualization?



Logistic Regression

- What if I plotted P(Y|X) vs X instead of Y vs X?
- But should P(Y|X) be a linear? Then what about extrapolating. Could it go below 0 and above 1?
- If we call P(Y|X) as π
- Maybe we can plot $\frac{\pi}{1-\pi}$ vs x (this allows the line to go from 0 to ∞)
- What about negative values. What if we plotted $\log\left(\frac{\pi}{1-\pi}\right)$ against X
- This is what we do; $\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X$
- $\pi(X) = \frac{\exp(\beta_0+\beta_1 X)}{1+\exp(\beta_0+\beta_1 X)}$
- Also, provides and ideal S-shaped curve

Logistic regression

- When there are more classes:

$$\log\left(\frac{\Pr(G=1|X=x)}{\Pr(G=K|X=x)}\right. = \beta_{10} + \beta_{11}x$$

$$\log\left(\frac{\Pr(G=2|X=x)}{\Pr(G=K|X=x)}\right. = \beta_{20} + \beta_{21}x$$

or

$$\Pr(G=m|X=x) = \frac{\exp(\beta_{m0}+\beta_{m1}X)}{1+\sum_{l=1}^{K-1}\exp(\beta_{l0}+\beta_{l1}X)}$$

$$\Pr(G=K|X=x) = \frac{1}{1+\sum_{l=1}^{K-1}\exp(\beta_{l0}+\beta_{l1}X)}$$

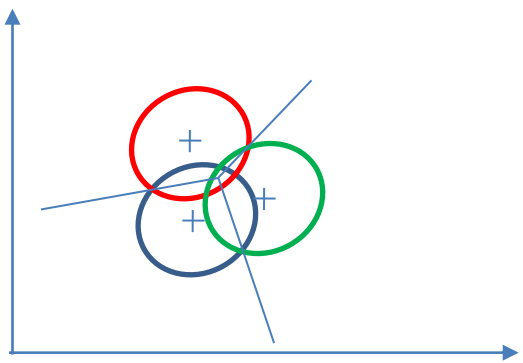
- What about the fitting?
  - Maximum Likelihood Estimation. How?
  - Maximize $l(\theta) = \prod_{i=1}^{N} P_{G_i}(x_i;\theta)$ or $\sum_{i=1}^{N} log P_{G_i}(x_i;\theta)$

Linear Discriminant Analysis

- How is it different from Logistic regression?
  - Logistic starts with an explicit relationship of Pr(Y=k|X=x)
  - LDA models the distribution of each class(k) in the X space, and then use Bayes theorem to get estimates of the form Pr(Y=k|X=x)
- Why do we need it? According to ISLR:
  - When classes are well separated
  - If n is small and the distribution of predictors is approximately Normal in each class
  - Popular when we have more than two classes
  - Also helps identify centroids

Linear Discriminant Analysis

- Uses probabilistic representation of each class
- In its base case each class is represented as a multivariate Gaussian (but it simplifies to other descriptions)
- Decision boundaries are build based on class conditional densities (this picture shown) as well as the prior probabilities
- Why do we need the priors?



Linear Discriminant Analysis

- $\Pr(G = k|X = x) = \frac{f_k(x)\pi_k}{\sum_{l=1}^{K} f_l(x)\pi_l}$

- Suppose we model each class density as multivariate Gaussian
- $f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T|\Sigma_k|^{-1}(x-\mu_k)}$
- What about π_k?
- Can we assume same covariance matrix?
- discriminant functions can be built are built on
- $\delta_k(x) = x^T\Sigma^{-1}\mu_k - {\mu_k}^T\Sigma^{-1}\mu_k + log\pi_k$; how do you get μ_k, π_k, Σ
- Σ tends to be weighted average across classes

Linear Discriminant Analysis

- To use this in a classifier between two classes(Decision boundaries)
- $log\frac{\Pr(G=k|X=x)}{\Pr(G=l|X=x)} = \log\left(\frac{f_k(x)}{f_l(x)}\right) + \log\left(\frac{\pi_k}{\pi_l}\right)$ which expands to
- Expanding and Simplifying
- $= \log\left(\frac{\pi_k}{\pi_l}\right) - \frac{1}{2}(\mu_k + \mu_l)^T\Sigma^{-1}(\mu_k - \mu_l) + x^T\Sigma^{-1}(\mu_k - \mu_l)$
- What about Quadratic Discriminant analysis and non linear boundries

ROC curves and contingency tables

- Wikipedia contributors. "Receiver operating characteristic." *Wikipedia, The Free Encyclopedia*. Wikipedia, The Free Encyclopedia, 17 Oct. 2018. Web. 24 Oct. 2018.:

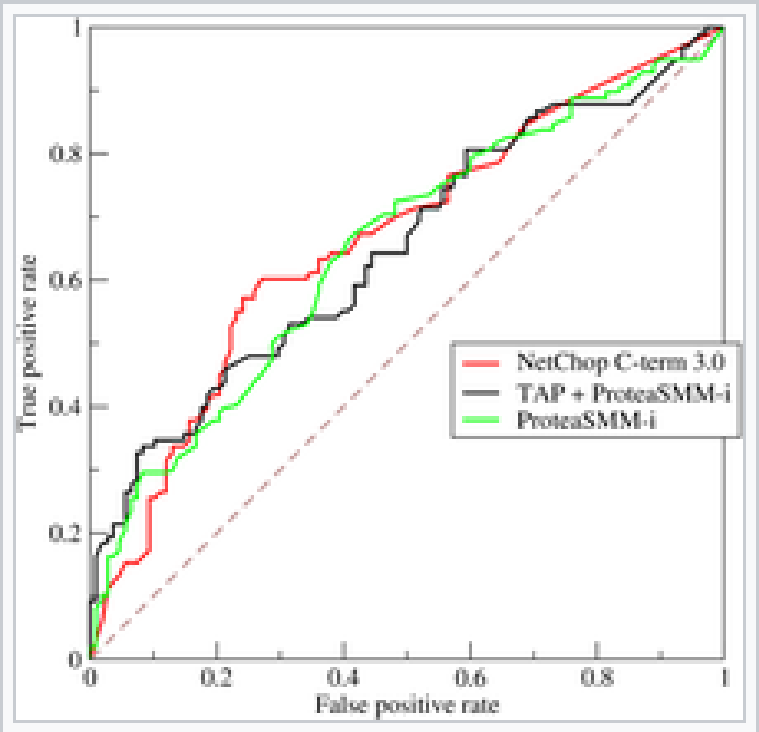
		True condition				
	Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR−}}$	F_1 score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	True negative rate (TNR), Specificity (SPC) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) = $\frac{\text{FNR}}{\text{TNR}}$		

ROC curves and contingency tables

- Accuracy = (TP+TN)/Total population
- TPR = Recall = sensitivity = TP/CP
- FPR = FP/CN

Some others

- Specificity = TN/CN
- Prevalence = CP/TP
- Precision = TP/(TP+FP)
- F1 = 2/((1/Recall)+(1/Precision))



Wikipedia contributors. "Receiver operating characteristic." *Wikipedia, The Free Encyclopedia*. Wikipedia, The Free Encyclopedia, 17 Oct. 2018. Web. 24 Oct. 2018.:

Support Vector Machines

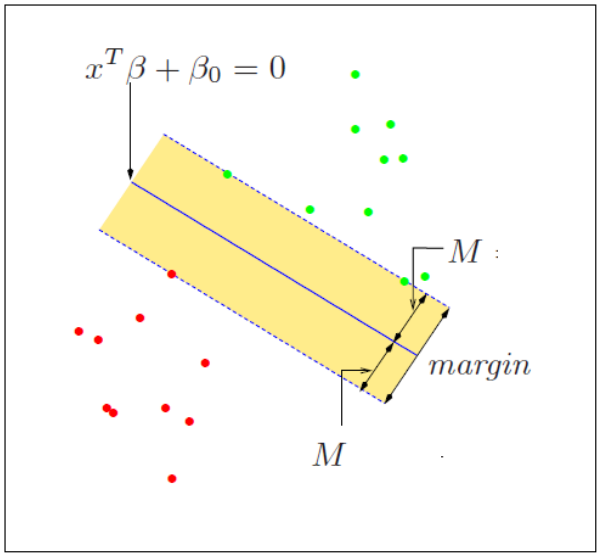
- Drawing Decision boundaries
- The <u>linearly separable classifier</u>

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

Or

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}) > 0$$



Hastie, Trevor, et al. "The elements of statistical learning: data mining, inference and prediction." *The Mathematical Intelligencer* 27.2 (2005): 83-85.

- But there are infinite such planes that satisfy this constraint
  - The need for a maximal margin
  - If $(x_i{}^T\beta + \beta_0) = 0$ defines a hyperplane then so does $k(x_i{}^T\beta + \beta_0) = 0$

Support Vector Machines

- So our objective is to:

$$\underset{\beta_0,\beta_1,\ldots,\beta_p}{\text{maximize}}\, M$$

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \ldots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n.$$

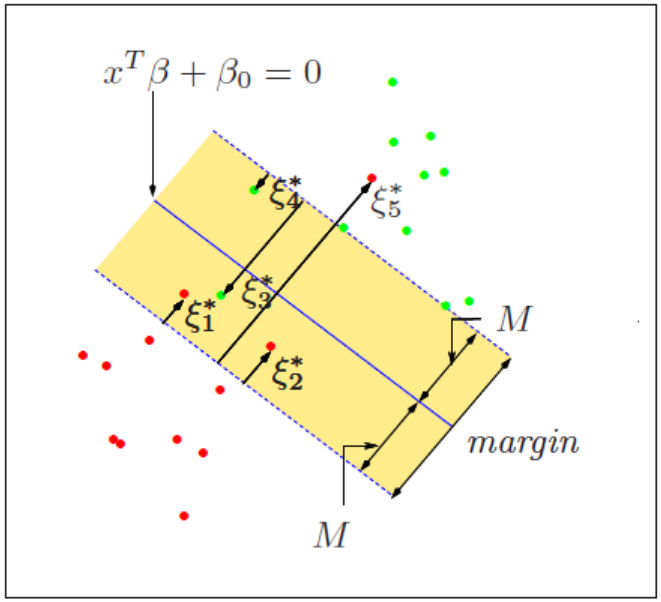
James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (Vol. 112, p. 18). New York: springer.

Support Vector Machines

- Extending this to the non-separable case. The <u>support vector classifier</u>
- Change the constraint to: $y_i({x_i}^T\beta + \beta_0) \geq M - \mathcal{E}_i$, where $\mathcal{E}_i$ for all values on the correct side is 0.

Or, more commonly

- Change the constraint to: $y_i({x_i}^T\beta + \beta_0) \geq M(1 - \mathcal{E}_i)$, where $\mathcal{E}_i$ for all values on the correct side is 0.
- With just this modification what would happen?
- We need to constrain the $\mathcal{E}_i$s through $\sum_{i=1}^{N} \mathcal{E}_i \leq C$
- Bias-Variance Dichotomy!



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Support Vector Machines

- Going from linear decision boundaries to non-linear ones: Same trick that we saw in regression to create more complex transformations of input variables.
- SVMs tap into this core idea and try to automate it with implicitly introducing new inputs (since that number can be very large)
- The $f(x) = x^T\beta + \beta_0$ is rewritten as some function of x; $f(x) = h(x)^T\beta + \beta_0$